# Template I

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## 1 How to use

Change the title, author, and footer email. Toggle solutions with `\setbool{sol}{false}`.

## 2 Gallery

**Theorem 2.1**

A finite graph admits an $\varepsilon$-regular partition with at most $M(\varepsilon)$ parts.

**Definition 2.1**

The edge density of $X, Y \subseteq V(G)$ is $d(X,Y) = e(X,Y)/(|X||Y|)$.

**Example 2.1**

Complete bipartite graphs are 0-regular pairs.

**Lemma 2.1**

Energy is monotone under refinement.

**Proposition 2.1**

Energy is a convex combination of squared densities.

**Corollary 2.1**

The tower-type bound cannot be replaced by a polynomial in $\varepsilon^{-1}$.

**Exercise 2.1**

Compute $d(X,Y)$ for a random bipartite graph.

**Proof.** Energy increases by at least $\varepsilon^5$ at each irregular step. □

**Solution.** Partition, refine, and stop when every pair is regular. □

**Remark.** Keep a single $\varepsilon$ for both size and density deviation. □

**Analysis.** The increment comes from the variance of a random edge density. □

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**Claim.** Empty blocks do not contribute to the energy. □

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